Technologies for Home Cage Monitoring in Preclinical Research

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Abstract

This chapter provides a comprehensive overview of technological solutions for home cage monitoring (HCM) in biomedical research. It outlines various monitoring needs and the corresponding technologies, including video-based systems, RFID, electromagnetic field technology, motion sensors, load cells, telemetry, metabolic monitoring, environmental sensors, and acoustic monitoring. The chapter discusses the

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technological overview, use and measurements, pros/cons, and specific considerations of each system. It also addresses future trends, combinations of systems, emerging technologies (such as AI-based computer vision), and aspects to consider when driving innovation in HCM technologies. The aim is to guide scientists in selecting appropriate tools and understanding the complexities of data collection and analysis in the HCM world.

Keywords

Technological solutions · Monitoring systems · Sensor integration · Behavioral analysis · Physiological monitoring

1 Introduction

Home-cage monitoring (HCM) systems are an integrated set of technologies that continuously and automatically observe and record animal behavior, physiology, and environmental features within an animal's home cage. Combining hardware (sensors and actuators) and software, HCM systems collect data with minimal human intervention. These systems ensure basic animal needs are met while enabling long-term monitoring, providing reliable, reproducible data for improved biomedical research.

In behavioral and physiological research, selecting the right toolset begins with a clear definition of the scientific question. Instead of focusing on the technologies themselves, researchers should first identify the type of monitoring required—e.g., locomotor activity, individual differentiation, physiological readouts, and environmental parameters—and then choose the appropriate methods to address their goals. Table 1 exemplifies this approach, categorizing available technologies based on the specific monitoring tasks they enable. By aligning experimental needs with the appropriate tools, this framework ensures that the data collected are both relevant and robust, streamlining the path from question to discovery.

HCM technologies offer a diverse array of tools tailored to specific experimental objectives, enabling detailed behavioral and physiological studies in laboratory animals. These tools range from video-based systems, capable of tracking individual or group behaviors with high accuracy and performing 3D pose reconstruction using multicamera setups, to RFID systems, which provide robust individual identification in group-housed conditions

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Table 1 Overview of HCM technologies categorized by monitoring needs, measured parameters, and compatible technologies

Type of monitoring need	Measurements needed (parameters)	Available technologies
Continuous cage tracking	Locomotion (circadian activity)	Video, RFID, beam-break, passive infrared sensors, capacitive electrostatic sensing, load cell, running wheel, telemetry implants, EMF
	Positional preference, social grouping	Video, RFID, capacitive electrostatic sensing
	Bouts of specific behaviors (sleeping, rearing, grooming, social interaction)	Video, 2 layers of IR sensors for rearing, load cells for scratching, EMF (for single-housed animals), acoustic monitoring (USV)
	Differentiation of individuals	Video, RFID
Recording events in the cage	Eating	Video, RFID, beam-break, passive infrared sensors, (to some extent) load cells
	Drinking	Video, capacitive electrostatic sensing
	Location in cage	Video, RFID, beam-break, passive infrared sensors, EMF
	Interactions with devices	Video, capacitive electrostatic sensing, beam-break, lever, touchscreen
Wearable / implantable devices	Bioelectric signals (e.g., ECG or EEG) *High sampling rates required*	Telemetry implants
	Biochemical signals (blood glucose), body temperature	Various: telemetry implants Temperature: IR camera, RFID implants
	Movement data	Accelerometers, gyroscopes
Environmental data	Temperature, humidity, light intensity, noise levels	Various sensors
Welfare	Biomarkers	EMF (urination), video (for posture analysis)

The table highlights available tools for continuous cage tracking, event-based recording, wearable sensing, environmental monitoring, and welfare assessment in laboratory animals. This list is not exhaustive

by detecting animals as they move through specific areas or interact with objects. Beam-break sensors and passive infrared sensors offer simple, event-based monitoring for activities like nose pokes or general locomotion, although they lack individual identification. Advanced methods like capacitive electrostatic sensing and load cells enable precise spatial tracking or unique measurements such as body weight, albeit with complex calibration and data processing requirements. Wearable/implantable devices like accelerometers and gyroscopes further extend monitoring capabilities, offering potential insights into fine-grained movement and equipment interactions. Table 1 organizes these technologies by their type of monitoring, guiding researchers to match their experimental needs (measurements needed), ensuring the available technologies match their scientific goals.

HCM technologies integrate various components to capture and analyze animal behavior in their natural environment. As summarized and simplified in Fig. 1, hardware

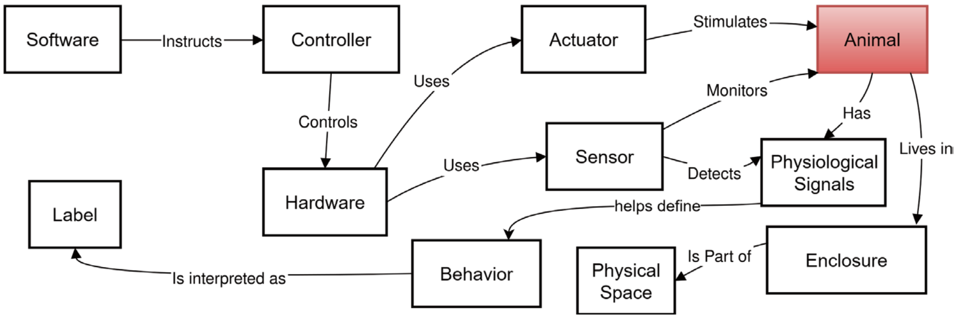

Fig. 1 HCM definition, adapted and simplified from TEATIME's Olog definition (Bains et al. 2025)https://www.cost-teatime.org/about/hcm-definition/. Specialized software controls hardware systems with sensors that monitor animal physiology and behavior. Actuators complement sensors to stimulate responses for dynamic experiments. Interpreted sensor data define labeled behaviors for analysis, yielding research insights. The system operates within an enclosure mimicking natural conditions, ensuring accurate tracking, stimulation, and interpretation for preclinical and behavioral studies

systems, controlled by specialized software, deploy sensors to monitor animals' physiological signals (e.g., ECG and EEG) and movements (e.g., locomotion and rearing). These sensors are often complemented by actuators, which interact with the animals to stimulate or influence specific responses, enabling more dynamic and precise experiments. The collected data are interpreted to define behaviors based on signals or movement patterns, which researchers label and analyze to derive experimental insights. The entire system operates within an enclosure (e.g., cage) designed to mimic the animal's natural living conditions, forming part of the broader physical space. This seamless interaction between sensors, actuators, and software ensures accurate tracking, stimulation, and behavioral interpretation, providing valuable data for preclinical and behavioral studies.

2 HCM Technologies

In this section, an overview of the technological landscape currently shaping the HCM field is provided. Rather than offering a comprehensive catalog, as was already organized by the COST ACTION TEATIME (CA20135) consortium (thebehaviourdatabase.org), we aim to highlight a representative selection of technologies—ranging from commercially available platforms to emerging open-source or experimental setups—that illustrate the diversity and evolution of approaches. Our analysis focuses on key dimensions relevant to both researchers and developers, including data output and scalability, accuracy and resolution, energy and computational efficiency, ease of use, cost, ethical considerations, modularity, and throughput capacity. While not exhaustive, this review strives to be both accurate and up to date, offering a structured framework to assess HCM technologies.

2.1 Video-Based Monitoring Systems

Among the various HCM technologies, video-based monitoring systems stand out for their ability to noninvasively capture a rich array of behavioral data. Furthermore, video systems can also directly monitor environmental variables, such as water or food availability and intake, flooding events, cage presence or absence in the rack, cage ID, and metadata (e.g., precise timestamps of light cycling or personnel entrances). This can greatly reduce the complexity of the sensor suites required for HCM solutions at scale.

2.1.1 Technology Overview

Video-based monitoring systems leverage standard video recording techniques alongside specialized methods such as infrared (IR, e.g., Kinect cameras) and thermal imaging (e.g., FLIR cameras) to monitor animals around the clock. Standard cameras capture high--resolution, color images during the day, while infrared cameras facilitate nocturnal monitoring (while still being fully functional in daylight), ensuring that observations continue seamlessly throughout the light–dark cycles. The typical configuration involves one (or more) cameras strategically mounted above (and/or around) the home cage. This setup is often enhanced by automated tracking and analysis software that employs computer vision and machine learning (ML) algorithms to quantify behaviors. Such algorithms are designed to identify and track individual animals, analyze their movements, and classify behavioral patterns. This technology not only enables the precise quantification of basic activities, such as locomotion and grooming but can also can be adapted to capture more complex social interactions and subtle postural changes, key aspects in preclinical research.

The main considerations when designing video monitoring systems relate to visibility. It is important to have high image clarity, no blind spots, and appropriate illumination (visible or IR) with as few reflections and shadows as possible. As home cages are typically required to be autoclavable, video monitoring systems are best placed outside the cage. In single-camera recording systems, this raises important geometric considerations to avoid blind spots, such as the viewing angle of the lens, relative position and size of the window, and location of occluding objects, like water bottles or food grids. Blind spots can be particularly challenging in cages with climbing enrichments, where ceiling surfaces and floor surfaces may need to be monitored simultaneously.

Because video systems produce very large amounts of data, the downstream processing and connectivity need to be designed into the setup. Where will storage and/or processing take place? Locally in the cage? On a rack-based server? In the cloud? What limits on the near-real-time availability of insights does this imply? How will video acquisition schedules be controlled? It is always tempting to aim for increasingly more data, but a video monitoring system that performs 24/7, high-resolution, 30 frame/s, full behavioral analysis can face very different computational hardware design bottlenecks from a system that performs scheduled analyses and occasional health and environment checks.

Real-time analysis also presents data bottleneck issues: video data compression is far more efficient when performed on sections of video rather than individual frames. Several models of camera popular in mouse behavioral analysis transmit video in an uncompressed format, which dramatically changes downstream hardware requirements when scaling to a whole-rack HCM system. In most HCM applications, this kind of ultralow video latency is not needed, and compression can be designed into the system.

2.1.2 Use and Measurements of Video Technologies

Video technology in HCM offers a noninvasive approach to measure a diverse range of behavioral and physiological parameters. Researchers utilize video recordings to assess the following:

- Locomotion and Activity Patterns: Detailed tracking of movement patterns provides insights into general activity levels, circadian rhythms, changes in locomotor behavior over time, and information on body postures (such as *rearing* or *climbing* behaviors).
- Social Interactions: Group-housed animals can be observed to study social hierarchies, mating behaviors, and aggression, all of which are crucial in behavioral studies.
- Grooming and Self-Care Behaviors: Video technology can quantify grooming frequency and duration, which are often indicators of stress or neurological conditions.
- Sleep and Resting States: Video monitoring, particularly with infrared technology, allows for accurate detection of sleep and resting without disturbing the animals.
- Eating and Drinking Behaviors.
- Physiological Parameters via Thermal Imaging: Thermal cameras can detect variations in body temperature, which may reflect underlying physiological changes such as fever, stress, or metabolic shifts.

In addition, advanced systems are exploring the integration of video data with other sensor modalities, potentially enabling the derivation of parameters such as respiratory rates or even subtle heart rate fluctuations when combined with high-resolution imaging and advanced analytical algorithms.

2.1.3 Keypoint Tracking Methods

Due to the difficulty of directly extracting measures from image data, the video analysis approach usually includes tracking the movement of keypoints of interest across the timespan of the video. This is done using background subtraction, or neural networks. Background subtraction methods for video analysis in HCM operate by comparing each video frame to a static reference frame, or background model, and identifying changes in pixel values. These changes are assumed to represent moving objects, ideally the animal. First applied to behavioral analysis in the late 1990s, this computationally efficient algorithm requires minimal setup, with static parameters like thresholds set once and applied across the entire dataset. Its predictability in terms of time investment is a key advantage. However, background subtraction alone is limited to single-animal tracking, typically tracking the center of mass of the animal's silhouette. It necessitates a significant visual distinction between the animal and the background and is highly susceptible to background changes, such as shifts in bedding. These limitations make it practically unsuitable for most complex home cage monitoring scenarios (Pennington et al. 2021).

Neural networks (NNs) offer a more advanced approach to video analysis, employing "black-box" models trained on researcher-labeled data. This iterative process involves labeling representative images from videos (typically 100–1000), training the NN, evaluating its performance, and then labeling additional images, often focusing on edge cases, before retraining. This continues until satisfactory performance is achieved. While this workflow has an unknown time investment and requires significant computational resources, including graphical processors (often found in gaming computers or institutional high-performance computing centers), a well-trained NN can achieve high accuracy in labeling diverse features, including specific body parts. The errors produced by NNs are usually pronounced and easily identifiable, and the variability in labeling between researchers can be quantified to ensure robust training data as well as a scale on which the model's accuracy can be gauged. Furthermore, NNs can be used for sophisticated tasks such as skeleton reconstruction, 3D pose estimation, and multianimal tracking (Pereira et al. 2022; Mathis et al. 2018; Lauer et al. 2022). See additional info in Sect. 3.5.

2.1.4 Pros of Video-Based Monitoring

One of the primary advantages of video technology in HCM is its ability to capture extensive and nuanced behavioral data without physical interference. Key benefits include

- Rich Behavioral Data: Video recordings provide a comprehensive visual record, allowing for detailed post hoc analysis of behavior. This is particularly useful for studying complex behaviors that are difficult to quantify with other sensors.
- Non-Invasive Monitoring: By relying on cameras instead of implanted devices or direct contact sensors, video systems minimize stress and discomfort, ensuring that the animals' natural behaviors are not altered by the monitoring process.
- Adaptability and Flexibility: Video systems can be easily integrated into various experimental setups. Whether it is a single camera for a small-scale study or a multicamera array for large group-housed studies, the technology scales according to experimental needs.
- Automated Analysis: Advances in machine learning and computer vision have led to robust software capable of automatically detecting and quantifying behaviors. This reduces manual labor and potential human error in data interpretation.

2.1.5 Cons and Challenges of Video-Based Monitoring

Despite the numerous advantages, video-based monitoring comes with its own set of challenges:

- Large Data Storage Requirements: Continuous video recording generates vast amounts of data, necessitating substantial storage capacity and efficient data management solutions.

- High Computational Demands: The processing and analysis of high-resolution video, particularly when using advanced ML algorithms, require significant computational power. This can be a bottleneck in studies involving long-term monitoring.
- Potential for Tracking Errors: Automated tracking systems may occasionally lose individual identities, especially in group-housed or environmentally enriched scenarios.
- Video Encoding Issues: High-quality video encoding is essential to preserve data fidelity, but it also increases computational overhead.

2.1.6 Future Developments

Looking ahead, the integration of video with other emerging technologies presents exciting possibilities. For instance, ongoing developments aim to incorporate physiological measurements such as respiratory patterns and body weight assessments from video data using computer vision (Guzman et al. 2024). Experimental setups are being designed to monitor not only adult rodents but also pups, thereby expanding the utility of video systems across different developmental stages (Lapp et al. 2023).

2.2 Systems Using RFID Technology

Radio frequency identification (RFID) uses radio waves for contactless data transmission, evolving from radar in the 1940s to wireless data exchange by the 1970s. It transmits information from a reader-powered transponder ("microchip") to a reader via radio waves, enabling the transmission of small amounts of data. Unlike other methods, RFID does not require a clear line of sight, allowing transponders to be embedded.

2.2.1 Technology Overview

There are three main types of RFID technology used today: low frequency (124–135 kHz), used for implantable devices like animal microchips due to its ability to penetrate tissue and water, which offers a read range of up to several meters. High frequency (13.56 MHz), suitable for vial labels in freezing conditions, enabling anti-collision reading in the short range (1–2 cm); however, it is unsuitable for implants. Ultrah-high-frequency (433–960 MHz) RFID, used in toll systems and retail security for its long range (up to 30 m) and active RFID systems with the highest range (up to 100 m) and fast data transmission, yet again not suitable for implantation or close contact.

For over three decades, LF RFID microchips have been utilized for animal identification, offering a permanent, reliable identification method. By implanting a transponder under the skin of laboratory animals and using compatible readers, researchers can achieve precise identification and tracking across various species. With microchips ranging from 1.2 mm to 2.1 mm in diameter and 6 mm to 13 mm in length, they are ideal for use in small animals such as mice and rats.

LF RFID systems typically consist of a reader equipped with wire coils that generate an electromagnetic signal. When a transponder comes into the range of the reader's signal, it absorbs energy from the electromagnetic field and modulates the signal to transmit stored data. The range and effectiveness of the system depend on factors such as the properties of the coil (diameter and number of windings), reader power, and the frequency of the signal. LF RFID microchips use either full duplex (FDX) or half duplex (HDX) communication methods. ISO 11784/85, more specifically FDX-B (134.2 kHz), is the most common communication standard in animal identification, offering an optimal read range and data transmission capabilities, making it ideal for laboratory animal tracking.

2.2.2 Use and Measurements of RFID Systems

Over the past decade, advancements in RFID technology have enabled the tracking of mice and rats in their home cage environments, opening new avenues for scientific exploration. One example is the study by researchers at Pfizer Worldwide Research, which combined RFID with home cage monitoring technology to track respiratory rates in laboratory animals (Winn et al. 2021). A reader plate was placed under the animal's home cage to track individual animals' data throughout the experiment. Another example is the Live Mouse Tracker (LMT, more details at https://micecraft.org/lmt/), which can automatically extract the (social) behavior of group-living mice by combining RFID and video data from an IR camera (de Chaumont et al. 2019; Huzard et al. 2022). Utilizing an implantable microchip and a specialized reader system allows for undisturbed monitoring of animals in their home cage. Previously, such evaluations relied solely on visual observation or required removing the animals from their cages, which introduced external stress and potential experimental variability.

The ISO 11784/85 standard provides a universal pet identification system for global compatibility, primarily for border crossing and lost pet recovery. Modern RFID technology extends this standard with increased memory to store additional data like temperature alongside identification (https://www.uidevices.com/laboratory-animal-temperature/). RFID technology's ability to monitor and track animals in real time has revolutionized behavioral studies and home cage monitoring. For instance, Sass et al. (2024) used RFID to track the thermogenic responses and metabolic rates of group-housed mice in response to neurokinin 2 receptor activation. RFID systems allowed researchers to monitor individual animals' activity levels and body temperature without the confounding effects of restraint or stress (Sass et al. 2024).

Another example of tracking animal movement and activity from cage to cage can be achieved using a tunnel reader. In the research study by Habedank et al., an RFID-based tracking system was used to determine the position of laboratory mice in a preference test (Habedank et al. 2022). The researchers described the process of using commercially available RFID components and an Arduino microcontroller to identify the mice as they pass through a tunnel in a housing system combined with two standard cages.

RFID has been used in seminaturalistic housing environments to collect detailed information on animal behavior while generating only small and manageable amounts of data. For example, a study using 29 RFID antennas tracked 20 mice, generating over 186,000 data points in 7 days while maintaining a compact total data volume of only 10.1 MB

(Mieske et al. 2021). These data can be analyzed using AI or specific scripts to monitor activity, temperature fluctuations, and behavioral patterns in real time, providing valuable insights into animal welfare and overall health.

2.2.3 Considerations with RFID Implants

RFID microchips should be implanted subcutaneously in the mid-scapular region (between the shoulder blades) using a caudal-to-cranial needle insertion to minimize infection and dislodging, typically resulting in a 1–2% fall-out rate for experienced researchers. Needle size guidelines vary based on microchip dimensions (e.g., ~12.5G for 2.1 mm × 12 mm). In postweaned animals, anesthesia is often unnecessary (acceptable in the US for >15 g mice with ≤15G needles and > 30 g rats with ≤12G needles), although European Directive 2010/63/EU classifies unanesthetized implantation as mild stress. Optimal RFID performance also depends on antenna size and positioning; tube antennas should match tube dimensions, while 3″ circular cage-side antennas offer a 3–4″ read range. Do-It-Yourself (DIY) antenna winding is technical and requires precise tuning to the reader, making consultation with RFID specialists advisable for proper functioning through inductance and capacitance adjustments.

For HCM, RFID can be used as a complementary tool alongside other technologies. The main benefit of RFID is its ability to track individual animals within a group-housed setting. While video systems can track a large number of animals using ear tags or other markers, they have limitations when animals are huddled together or out of the camera's view. RFID overcomes these challenges by detecting animals through bedding, nonmetallic enrichment devices, and water, making it a reliable identification method. When combined with temperature-sensing microchips, RFID also enables continuous animal welfare monitoring, as temperature is a key indicator of health and stress. Recent advances are pushing the development of newer implants combining various sensors (thermometer, accelerometer, and photoplethysmograph) within the RFID chip and allowing the recording of a wider range of behavioral, as well as health parameters (https://micetracking.com/). New data open new fields of research in medicine, pharmacology, and animal well-being.

2.2.4 Challenges, Limitations, and Future of RFID Technology

While RFID technology offers significant benefits, it also presents certain challenges, such as imaging interference and anti-collision signal interference. First, although most RFID microchips are compatible with MRI, X-ray, NMR, or body composition analysis, they can produce imaging artifacts. Researchers should assess the impact on their specific area of interest to ensure that image quality remains uncompromised. Second, when two or more microchips are present within the same antenna field, the reader may scan microchips A, B, or neither during a given scan. This occurs due to the nature of the LF RFID signals and how the reader interprets individual readings. When mice are huddled together, the reader may register one animal or none. To enhance data reliability, combining RFID monitoring for identification and temperature tracking with complementary methods such as video analysis can provide a powerful research solution.

RFID technology has proven to be an invaluable tool for tracking and identifying laboratory animals, providing precise and reliable data without the need for visual contact. Advances in RFID capabilities and applications in animal research continue to grow, enabling innovative methodologies for studying behavior, health, and welfare. By selecting appropriate equipment, optimizing implantation techniques, and adhering to ethical standards, researchers can fully leverage RFID technology while prioritizing animal welfare.

Combining RFID technology with additional sensors represents a significant advancement, enabling the simultaneous and continuous collection of diverse data. This integration in a single RFID implant ensures multiple measurements, highly repeatable experiments, and strong correlation with external factors (e.g., day/night cycle and temperature) and activities (e.g., sleeping and eating). For instance, continuous photoplethysmography (PPG) measurements allow the calculation of physiological parameters like heart rate, blood pressure, respiratory rate, and blood oxygen saturation using digital signal processing (DSP) and machine learning (ML) techniques. Furthermore, the integration of chemistry-free sensors into a surgery-free RFID implant provides an ethical and cost-effective approach compared to traditional methods (https://micetracking.com/).

2.3 Electromagnetic Field Technology

Electromagnetic field (EMF) technology has emerged as a compelling solution for HCM, offering a noninvasive, continuous, and scalable method to capture various animal behavioral and physiological parameters. By integrating EMF sensors within the cage setup, systems such as Tecniplast's DVC® enable researchers to monitor group-housed animals in real time, collecting group data on activity levels and indicators like rest/wake bouts and polyuria patterns (Voikar and Gaburro 2020; Brachs et al. 2025).

2.3.1 Technology Overview

EMF-based HCM systems utilize an array of magnetic or capacitive sensors placed underneath or along the walls of standard animal cages. These sensors detect subtle perturbations in the local electromagnetic field caused by an animal's movement or presence. When an animal moves across the cage floor, its body—characterized by high water content—alters the local dielectric properties, modifying the field strength and distribution. The sensor outputs, typically voltage fluctuations, are recorded and processed to infer movement and position. For instance, the DVC® system from Tecniplast employs multiple electrodes to monitor changes in conductance as animals traverse the cage, providing quantitative measurements of activity and spatial positioning (Iannello 2019).

2.3.2 Applications and Measured Parameters

The EMF technology in HCM is designed to capture a range of measurements with minimal human intervention. Key parameters include

- Activity Levels: Continuous tracking of general locomotion allows researchers to assess patterns such as distance traveled and average speed.
- Social Interactions: Group-housed settings benefit from the ability to record interactions such as fights or aggressive encounters with minimal precision since the system cannot know which animals are implicated in this magnetic field fluctuation.
- Sleep Patterns: Changes in activity can serve as a proxy for sleep, as periods of inactivity during the dark phase of the circadian cycle often indicate rest.
- Wetness and Urination Events: Variations in the electromagnetic signal can also reveal the presence of fluids such as urine, which may indicate both normal physiology and recovery postsurgery, or indicate flooding of the cage.
- Environmental Room Conditions: Real-time conditions (humidity, temperature, human presence, noise, and vibration) can be monitored with a rack-based tool (see Sect. 2.8).

By providing data on these aspects, the DVC system supports refined phenotyping and welfare assessments without the need for continuous human observation.

2.3.3 Noninvasive and Continuous Monitoring

EMF technology offers noninvasive, 24/7 monitoring, minimizing stress and complications. Its low data storage needs (4 MB/cage/day with 4 Hz sampling) enable long-term studies and automated, continuous data collection, crucial for observing natural behaviors in group-housed animals while minimizing human interaction.

2.3.4 Group Housed Data Collection and Social Interactions

EMF-based monitoring tracks group-housed animals in ventilated rack cages, monitoring over 10,000 cages simultaneously without disturbance. It provides a comprehensive activity profile for each cage, revealing social dynamics like aggressive encounters and detecting subtle behavioral shifts (e.g., stress indicators from increased daytime movement in nocturnal species). This group-level data, when correlated with other parameters, offers insights into animal welfare. However, it lacks individual animal data or precise social dynamic details.

2.3.5 Pros of EMF

- Continuous, Real-Time Monitoring: EMF sensors provide ongoing surveillance, capturing rapid behavioral changes without gaps in data collection.
- Low Data Storage Requirements: The simplicity of the data (voltage fluctuations) means that even long-term recordings produce manageable data volumes.
- Suitability for Group Housing: EMF systems can monitor multiple animals simultaneously, preserving the social environment and reducing handling stress.

- Non-Invasive Measurement: avoiding surgical interventions improves welfare outcomes.
- Versatility in Measurement: EMF systems can capture data on rest/wake bouts and even detect wetness, which can be critical for monitoring the onset of hyperglycemia (Brachs et al. 2025).
- Ideal for combination with other technologies.

2.3.6 Cons and Limitations of EMF

- Limited Behavioral Resolution: While EMF sensors accurately record gross activity, they typically lack the detailed resolution required to distinguish fine-grained behaviors (e.g., grooming or subtle postural changes).
- Potential for Signal Interference: External electromagnetic noise or overlapping signals from closely interacting animals might lead to data artifacts. This can be particularly challenging in settings with high ambient interference or when cages are not properly shielded.
- Need for Specialized Cage Setups: Cages must be compatible with the sensor arrays to maximize the effectiveness of EMF sensors. Retrofitting existing cages might require modifications to ensure proper sensor placement and signal capture.
- Group Data Aggregation: Although group monitoring is an advantage, it also means that individual behavior is sometimes obscured, limiting the capacity for individual-level analysis in social housing scenarios.

2.3.7 Comparison with Other Technologies

Compared to video-based monitoring, which can offer detailed behavioral insights through advanced computer vision techniques, EMF-based systems prioritize high scalability and low invasiveness. Video systems tend to generate large amounts of data and require significant computational resources to process high-resolution images continuously. In contrast, EMF systems produce simpler data that are easier to store and analyze, making them more practical for large-scale, high-throughput studies. Similarly, while RFID systems excel at individual identification and tracking, they typically require the implantation of chips, which is invasive and may affect animal welfare. By operating externally, EMF technology avoids these drawbacks while still delivering valuable group-level data.

2.3.8 Perspectives and Conclusion

Looking ahead, improvements in sensor design, data processing algorithms, and refined analysis could help overcome some current limitations of EMF-based HCM systems. For instance, integrating machine learning approaches to deconvolve overlapping signals better could enhance behavioral resolution, enabling the system to differentiate between complex activities. Advances in sensor miniaturization and shielding could reduce the impact of external electromagnetic interference, improving data accuracy even in challenging environments. Additionally, combining EMF data with other noninvasive monitoring technologies—such as videos or microphones—could provide a more comprehensive understanding of animal behavior. Such systems might leverage the strengths of each method: EMF for continuous, low-data monitoring and video for detailed behavioral classification. This integrative approach holds promise for future research to refine phenotyping methods and enhance experimental reproducibility. Finally, improving scientists' ability to discover insights from existing datasets could be achieved by validating and refining EMF algorithms using AI models trained on data from systems that integrate multiple technologies.

EMF technology enables continuous, noninvasive home cage monitoring of activity, some social interactions, sleep, and urination. While less detailed than video tracking, its low data storage requirements and suitability for group housing make it valuable in HCM. Its scalability and easy integration benefit researchers designing large-scale, high-throughput studies. Ongoing development aims for improved animal welfare and research reliability. Despite limitations in detailed behavioral analysis and potential interference, EMF's noninvasiveness and scalability make it a preferred preclinical method.

2.4 (Other) Motion-Sensing Technologies

Alongside EMF (and, in some uses, RFID technology), implantable accelerometers, passive infrared (PIR) sensors, and infrared beam break systems represent fundamental technologies used in HCM systems to quantify locomotor activity, assess circadian rhythms, and broadly capture movement-related behaviors in laboratory animals. These tools are especially valued in research settings where minimizing human interaction is essential for reducing stress and experimental artifacts while ensuring that animals' daily activities and behaviors are continuously monitored (Goulding et al. 2008). Similarly to EMFs, implantable accelerometers' complex raw output is attractive as it could be used to infer occurrences of various specific behaviors. On the other hand, PIR and beam break sensors have simple data outputs, but their simplicity and cost make them attractive options. Motion, proximity, or presence sensing using radar-like time-of-flight measurements (ultrasound or microwaves) are not present in HCM systems. Ultrasonic sensors are unsuitable due to rodent sensitivity. Microwave sensors, while having similar drawbacks to infrared sensors, also have higher complexity and cost, explaining their absence. Motion sensing can also monitor voluntary activity like running wheel usage in home cages.

2.4.1 Implantable Accelerometers in HCM

Accelerometers have been designed to detect and record subtle movements of the animal (S. Venkatraman et al. 2010). These sensors are embedded within implantable telemetry devices and provide high temporal resolution, making them ideal for studies where precise quantification of movement patterns is necessary (Ouyang et al. 2024). These devices capture detailed acceleration and orientation data across multiple axes, providing researchers with comprehensive activity profiles. High sampling rates (up to 1 kHz) record even minor movements, enabling accurate activity assessments over various timescales.

Accelerometer data are used to evaluate circadian rhythms by tracking patterns of activity and rest across the day–night cycle. In addition, they provide information about gross locomotor activity—essential for studies in behavioral neuroscience and physiology where changes in movement can be indicative of underlying conditions or responses to interventions (Seibenhener and Wooten 2015; Crawley 2007, 2023). Because the data generated by these sensors are relatively compact, storage and data processing demands are significantly reduced compared to more data-intensive methods like continuous video monitoring. This makes accelerometer-based HCM systems efficient and cost-effective for long-term studies.

Implantable accelerometers minimally impact animal behavior, allowing for natural movement recording after surgical placement. Although invasive, the procedure yields reliable, long-term movement data. Calibration is necessary for accurate acceleration reflection, and long-term studies must consider battery type (batteryless or rechargeable).

2.4.2 Infrared Beam Break Systems in HCM

Infrared (IR) beam break systems monitor animal movement via IR beam interruptions in a cage, offering basic 3D data on locomotion and rearing. Their advantage lies in continuous, high-sensitivity monitoring with minimal data storage, suitable for long-term studies where video is impractical. These systems integrate well with other technologies, enhancing HCM implementations.

IR beam break systems have limitations, notably their inability to provide detailed behavioral information. They detect movement but struggle to differentiate specific behaviors beyond beam interruptions. Early systems struggled with this, but Columbus Instruments later developed a system using microprocessors to classify beam breaks as ambulatory or stereotypic based on interruption sequences (Czekajewski et al. 1982). Initially, data collection was limited, but systems with integrated computers enhanced data output and enabled automated scoring (Sulaiman et al. 1986). Using a grid arrangement, animal position can be mapped using Cartesian coordinates, and based on beam break patterns, states like resting, sleeping (Pack et al. 2007), genotypic, and ambulatory (Kelley 1993) can be classified.

Locomotor movement is the movement from one position to another (Kelley 1993). Sometimes, stereotypic movement is scored incorrectly as locomotor movement, and with sophisticated software, “stereotypic filtering” can be employed. Software-based stereotypic filtering uses an imaginary fence to classify movements. Movements within the fence are stereotypic. If the subject’s centroid exits, the state becomes ambulatory, and the fence is erased and redrawn upon stopping. Abrupt ambulation from rest might initially classify steps within the fence as stereotypic, even if the path becomes ambulatory. Short periods of stereotypic movement immediately preceding ambulation can be reclassified as ambulatory if shorter than a set threshold.

Today, more data tools are becoming available for actigrams, heatmaps, and automated sleep scoring (Keenan et al. 2020). While IR beams have become excellent at recording locomotion, they still remain limited in two ways: animals must be singly housed, and the sensors lack sensitivity to further classify stereotypic movements into specific behaviors (grooming, feeding, or seizures).

2.4.3 Passive Infrared Sensors in HCM

Unlike beam break sensors, passive infrared (PIR) sensors do not emit infrared light—they are sensitive to the rodent's own infrared emissions. Usually, such a sensing element is part of a module with a specialized integrated circuit, which detects, amplifies, and thresholds the signal, finally outputting a logical 1 when the threshold is crossed, i.e., movement is detected. Such data require little storage space and compresses well.

Owing to the PIR sensor modules' simplicity, low cost (less than 20€ for a calibrated, industrial variant, 1€ for a hobbyist variant), and simple data output, this approach is especially attractive for DIY HCM, and Chap. 9 contains full instructions for building a simple prototype.

While PIR and beam break sensors have similar pros and cons, PIR sensors are better for larger areas and easier to install, often requiring just one sensor per cage for general activity. Beam break sensors need a horizontal beam, which can be obstructed by bedding or enrichment items. However, beam breaks excel at detecting presence at specific points, especially in small spaces like shelter or nosepoke openings.

2.4.4 Future Directions

Core sensing technologies like PIR, beam break sensors, and accelerometers have reached a maturity level limiting further innovation in the sensors themselves. Future advancements will likely stem from applying computational methods, such as machine learning for behavioral biomarker extraction from raw sensor data (as Tecniplast does with DVC system) and multimodal data integration. Innovation is shifting toward data fusion, advanced modeling, and behavioral inference algorithms.

2.5 Load Cells and Pressure Sensors

HCM systems offer a nonintrusive, continuous way to observe and record animal behavior without disturbing the animals. In this context, load cells and pressure sensors play a crucial role by capturing subtle weight shifts and changes in weight distribution that may reflect various behaviors. These sensors, typically placed under the cage or integrated into its floor, provide real-time data on locomotion and activity-related behaviors, such as scratching bouts (LABORAS system from Metris) — behaviors that are often challenging to capture with video-based systems alone.

2.5.1 Technical Overview and Operating Principles

Load cells and pressure sensors, integrated into cage floors, use strain gauge technology to detect minute weight or force changes as electrical signals. This noncontact method allows for the unobtrusive, continuous monitoring of animal behavior in a natural setting, detecting subtle weight distribution shifts for detailed analysis, although the exact cause of these variations may not always be identifiable.

2.5.2 Applications in Monitoring Behavioral and Physiological Parameters

Load cells and pressure sensors have a broad range of applications in HCM systems:

- Activity Levels: These sensors continuously track weight shifts, providing precise data on animal locomotion, including movement patterns, rest, and activity bursts, offering insights into circadian rhythms and behavioral changes.
- Body Weight Estimation: Integrated systems can estimate overall body weight from continuous data, reducing stressful handling and weighing (e.g., TSE AnimalGate).
- Detection of Fine Motor Behaviors: Advanced calibration and AI algorithms enable sensors to detect behaviors like scratching, often difficult to validate via video. This is valuable in studies where grooming or scratching indicates stress or neurological conditions (Huzard et al. 2024).

These sensors monitor group-housed animals, providing a reliable overview of group dynamics and activity trends through aggregated data, although individual identification may require additional technology.

2.5.3 Pros

Several key benefits make load cells and pressure sensors attractive for HCMs:

- Non-Intrusive and Continuous Monitoring: These contactless sensors allow for noninterfering behavioral recording, reducing stress variability and improving data reliability.
- Simplicity and Ease of Setup: Load cells easily integrate into existing cage systems, requiring minimal structural changes, making them a cost-effective and scalable solution for large studies.
- Versatility in Data Collection: Load cells offer valuable insights into behaviors like feeding, drinking, and scratching, making them versatile for diverse research, from metabolic studies to behavioral neuroscience.

2.5.4 Cons

Despite their many advantages, the use of load cells and pressure sensors comes with several caveats:

- Sensitivity to Environmental Vibrations: These systems are highly sensitive to vibrations, with environmental movements or cage disturbances causing data noise. Researchers must isolate the setup and use signal-processing algorithms to filter vibrations.

- Potential for Inaccurate Measurements: Inconsistent bedding or cage movement can skew weight measurements by altering sensor pressure. Regular calibration and correction factors are crucial to address this.
- Calibration and Data Processing: Accurate initial calibration and robust data systems are essential for reliable weight change monitoring from continuous sensor data, avoiding external interference and data loss.
- Integration with Other Modalities: Load cells are best combined with other sensors (e.g., video tracking, RFID, and telemetry) for comprehensive analysis, increasing complexity but improving behavioral interpretations.
- Group Data Aggregation: While these sensors can effectively estimate bulk activity and behavioral patterns in group-housed environments, it is difficult to discern readings of individual animals in these conditions.

2.5.5 Better Load Cell Data and Practical Examples

Recent advancements in sensor technology have led to improved load cell designs with enhanced sensitivity and better vibration isolation. A system like LABORAS has demonstrated the capability to not only monitor general activity levels but also detect specific behaviors that were previously challenging to measure, such as scratching (LABORAS system from Metris). Additionally, PASTA (Platform for Acoustic STArtle)—despite not being a full HCM system—is an open source example that demonstrates the ease of interfacing with load cells in a DIY setup (D. Virag et al. 2021). In the PASTA proof-of-concept experiment, periodic breathing as a result of the startle response was detected in video-validated PASTA output.

Integrating load cell data with telemetry or video tracking enhances understanding of animal behavior by correlating physiological and behavioral observations. This multimodal approach offers a holistic view, reducing reliance on single measurements and improving data accuracy. For example, combining telemetry's cardiovascular data with load cells' physical activity insights creates a more complete picture of an animal's well-being.

2.6 Telemetry Systems

2.6.1 Advantages of Telemetry in Research

Implantable telemetry is the most humane technique for collecting physiological signals from undisturbed, fully conscious, and freely moving laboratory animals (Kramer 2001) in different behavioral settings (Huzard et al. 2019). This wireless technology refines physiological monitoring in biomedical research since animal experiments are performed under stress-free conditions (B. P. Brockway et al. 1992), including home cage environments. As a result, experimental artifacts and stress-induced and intersubject variability are minimized, leading to increased accuracy, reliability, and predictive value of the data relative to studies using anesthetized or restrained animals (Kramer and Kinter 2003). Through refinement, telemetry helps to reduce the number of animals needed per study (Kinter 1996). Furthermore, telemetry assists researchers in identifying essential operational and experimental aspects to improve animal welfare (Toth et al. 2015; Meijer 2006). Telemetry implantation requires invasive surgery by skilled professionals following strict aseptic and ethical guidelines to minimize complications and distress. Training, standardized protocols, and refined techniques are crucial for scientific integrity and animal welfare.

2.6.2 Components and Functionality of Telemetry Systems

Telemetry uses implantable sensors to collect and transmit animal physiological signals to a remote transceiver. This transceiver then relays the data to software that saves raw data and generates derived parameters based on user instructions (Kramer 2001). Telemetry systems monitor multiple rodents simultaneously. A typical rodent study may utilize six to ten animals, but telemetry platforms are scalable and can accommodate more test subjects for larger-scale studies (Pedraza et al. 2025). The implantable sensors (or transmitters) are the key to conferring a high degree of modularity and versatility to the technique. Telemetry transmitters offer varying battery lives (1–12 months) and can measure signals like biopotentials, pressure, and blood glucose (Kramer et al. 2021). Transmitter type and surgical methods determine study length and which physiological signals (e.g., ECG, BP, EEG, EMG, glucose, temperature, and activity) can be recorded.

Cardiovascular evaluations via telemetry are accomplished by placing the body of the transmitter intraperitoneally or subcutaneously. For ECG measurements in rodents, the positive and negative electrodes are placed subcutaneously (Gkrouzoudi et al. 2023; Sgoifo et al. 1996). In this setting, proper electrode positioning is critical to acquiring quality ECG data from mice or rats. For BP recordings in mice, the less invasive method is placing the BP catheter into the carotid artery (Mills 2000). In contrast, surgical approaches using the abdominal aorta or femoral artery are more suitable for rats (Mattes and Lemmer 1991). The pressure-sensing technology also permits the assessment of left ventricular pressure, an indicator of cardiac contractility. Common cardiovascular investigations utilizing telemetry in rodents include heart rate variability analysis, physiological and mechanistic examinations, disease modeling, pharmacological testing, and circadian rhythms.

On the other hand, the typical use of rodent telemetry in neuroscience research includes seizure and sleep studies, neuropharmacology, phenotyping, and chronobiology (Kramer et al. 2021). In this regard, the most assessed physiological signals in rodents are EEG and EMG. Electrodes are attached to the skull for EEG measurements, with the leads placed epidurally or connected to deep electrodes (Lundt 2016). For combined EEG and EMG recordings, another set of electrodes is sutured to the neck muscles to obtain EMG signals. The body of the transmitter is generally placed subcutaneously.

2.6.3 Other Physiological Parameters Measured Via Telemetry

Another widespread practice of rodents instrumented with telemetry transmitters is to monitor body temperature and activity, crucial for pharmacology, infectious disease, and behavior studies. Continuous glucose telemetry improves welfare (Brockway 2015) via glucose oxidase enzyme catheters, aiding in understanding homeostasis and biomarker evaluation. Innovative telemetry applications include deriving respiratory rates from EMG or pressure signals and measuring pressures like intracranial, arterial, or ventricular. Telemetry integrates with phenotyping (Sommer 2005), fear conditioning (Gaburro 2011), and metabolic cages (Gaigé 2014). However, excessive multiparameter data collection may negate home cage advantages.

2.6.4 Challenges, Considerations, and Future Developments of Telemetry Implementation

Telemetry, an invasive method requiring surgical transmitter implantation, offers benefits but presents implementation challenges. Minimizing discomfort and complications hinges on appropriate animal body weight and transmitter type, alongside skilled surgical staff for successful implantation and reduced postoperative issues. Following proper care and recovery, rodents tolerate transmitters well, enabling longitudinal studies (Baumans 2001). Second, telemetry enables scheduled or continuous data acquisition, usually at sampling rates between 600 and 1000 Hz. Raw data from telemetry signals can generate large files. For example, a rodent experiment that collects data continuously for 7 days is estimated to render approximately 1.5 gigabytes of information per subject. Reasonable computing capacity (8 Gb RAM minimum, Intel® Core™ i7) and sufficient disk drive storage space (250 Gb) are needed, and competent software modules that automate data analysis and event detection are beneficial, especially for long-term studies. Last, the acquisition of a telemetry system involves strategic budgeting, capital allocation, and planning and running costs for transmitters. Nevertheless, the cost is generally wholly recovered through reduced animal use (Kramer et al. 2021). Additionally, users often leverage the versatility of telemetry to provide research services to maximize returns on the investment.

Telemetry technology continues to progress. Modern telemetry platforms count on group housing capabilities to monitor groups of rodents in their home cages (Martin 2024) and extended transmission ranges to accommodate high-density caging environments with minimal signal interference. Further advancements could witness the development of new sensors to measure additional parameters, cloud-based data collection, smaller transmitters, and longer battery life.

2.7 Metabolic Monitoring Systems

Metabolic technology, particularly indirect calorimetry, is a crucial part of home cage monitoring for studying energy balance, obesity, and metabolic health. It measures oxygen consumption and carbon dioxide production to determine metabolic rate, energy expenditure, and homeostasis. Additional sensors track activity and food intake, providing context without stressing the animals. These measurements also reveal diurnal changes in RER, indicating whether carbohydrates (RER closer to 1.0) or fats (RER nearer 0.7) are being metabolized (Deuster and Heled 2008). This level of detail is crucial for studies examining metabolic flexibility, the impact of dietary interventions, and the progression of metabolic disorders. However, indirect calorimetry requires the isolation of animals, and it would be challenging to adapt it to the group-housed context, as all subjects in the cage would be judged as one biomass.

2.7.1 Integration with Activity Data for Comprehensive Metabolic Profiling

Physical activity significantly contributes to energy expenditure. HCM technology offers many options for tracking home-cage locomotor activity, with IR beam activity meters being the gold standard for accuracy, scalability, and compact size. Analyzing metabolism correlated to activity is nuanced due to the causal relationship between activity, behavior, and metabolism. Modern metabolic phenotyping systems can align activity events with increased energy expenditure, and specialized software can analyze TEE differences based on activity levels, enhancing data robustness and experimental repeatability (Škop et al. 2023).

2.7.2 Integration with Feeding Data for Comprehensive Metabolic Profiling

Modern metabolic HCM systems integrate calorimetric data with feeding and water intake records for comprehensive analyses, correlating energy expenditure with nutritional intake. This integration is vital for understanding metabolism, behavior, and environmental factors, enabling energy balance calculations crucial for obesity research and metabolic syndrome studies (Rubio et al. 2023).

HCM systems use specialized software to analyze real-time metabolic data, providing immediate insights and enhancing the predictive value of experimental data by identifying patterns like circadian rhythms or dietary responses.

2.7.3 Pros of Metabolic Monitoring in HCM

- Rich Data Collection: The ability to monitor oxygen consumption, carbon dioxide production, and feeding behavior simultaneously provides a multidimensional view of an animal's metabolic state. This is crucial for studies focused on energy balance and the pathophysiology of obesity.
- Reduced Stress Artifacts: While isolation for indirect calorimetry can cause stress, home-cage metabolic sensors reduce this by shortening separation times. Future methods may enable group measurements using stable isotopes but currently face technical and cost challenges.

- Enhanced Experimental Relevance: Continuous and long-term monitoring allows for the detection of subtle physiological changes over time, making it possible to assess the effects of chronic interventions or the progression of metabolic disorders.

2.7.4 Cons of Metabolic Monitoring in HCM

- Complex Setup and Maintenance: Implementing indirect calorimetry in a home-cage setting involves sophisticated hardware and software integration. The system must be meticulously calibrated to account for environmental factors, such as bedding materials, cage design, and ambient temperature.
- Potential Housing Limitations: While the integration of metabolic monitoring into HCM systems is advancing, many setups still require individual housing to ensure precise measurements. This can limit the study of social behaviors and may necessitate modifications to standard HCM configurations.
- Data Management Challenges: Continuous data acquisition results in large datasets that require robust computational resources and specialized analysis modules to process and interpret effectively.

2.7.5 Commercial Examples and Innovations

Several commercial systems have been developed to meet the demands of metabolic monitoring in home-cage environments. Examples include the Sable Promotion, TSE PhenoMaster, and Oxymax-CLAMS (Comprehensive Lab Animal Monitoring system). These platforms are designed to integrate indirect calorimetry with additional behavioral and environmental monitoring tools. The CLAMS system, for instance, has been successfully used in collaborative research to evaluate the interplay between bedding, metabolic rates, and feeding behavior, providing a model for comprehensive metabolic profiling in a naturalistic setting (Speakman 2013; Eynaud et al. 2024; Hu et al. 2023; Škop et al. 2020).

Additionally, plethysmography—while conceptually attractive for respiratory monitoring—faces significant practical limitations in a full-volume home-cage setting. Accurately recording tidal volume becomes infeasible once the cage airspace exceeds a few liters, and although the breathing rate can still be tracked, the signal-to-noise ratio is poor. In fact, Columbus Instrument is currently the only commercial provider to offer respiratory frequency in a home-cage context, but it is planned to phase it out due to these technical challenges. Previously, smaller CLAMS "center-feeder" cages (with roughly half the internal volume) yielded more reliable plethysmographic signals, but those formats are no longer in widespread use.

2.7.6 Enriching HCM with Advanced Metabolic Insights

Recent advances have greatly improved HCM metabolic monitoring. Sensor technology offers enhanced sensitivity and accuracy, with cloud-based data and real-time analytics enabling remote monitoring. Algorithms and machine learning now automatically classify metabolic events, linking them to behavior or environment for deeper physiological insights. Miniaturized sensors reduce invasiveness and allow for group housing integration, with ongoing research addressing accuracy challenges to create more versatile and scalable HCM systems.

2.7.7 Future Perspectives

The future of metabolic technology in home-cage monitoring promises less invasive, more precise, and broader data collection. Wireless, battery-free sensors could enable continuous monitoring, akin to advancements in physiological telemetry. Integrating metabolic data with video-based behavioral analysis and environmental sensing will create multiparametric platforms, streamlining experiments, and reducing animal numbers per study by providing comprehensive datasets.

2.7.8 Conclusion

Integrating metabolic technology, particularly indirect calorimetry (VO_2, VCO_2, and RER) and feeding data, into home-cage monitoring advances biomedical research. Despite setup and data management challenges, the benefits of enhanced experimental relevance and reduced stress make it invaluable for studying metabolism, obesity, and animal welfare. Technological innovations and multimodal integration will ensure that metabolic HCM systems play a key role in future preclinical research.

2.8 Environmental Monitoring

Environmental factors like illumination, humidity, and temperature significantly impact rodent behavior and physiology yet are often overlooked in studies. Broadly, there are three options for environmental monitoring systems. First, there is a commercial, lab-oriented system, e.g., Pallidus.io. This system is composed of battery-powered, cage top-mounted units that send activity, humidity, and temperature data to a real-time cloud dashboard. Second, there is a segment of the home automation market for wireless, networked sensors for home air quality data, such as Aqara, AirGradient, Apollo AIR-1, or Sonoff, which could ostensibly be repurposed for laboratory use. Finally, there are lab-developed open-source solutions, such as the MIROSLAV device with the EnviroSLAV addition to the main unit, which also provides real-time wireless data transmission to one or multiple logging devices (Virag et al. 2025). While the EnviroSLAV addition was originally developed to monitor habitat conditions, attaching the main MIROSLAV unit to cage tops is feasible. For example, recently, a group developed a real-time ambient sensing tool, MyVivarium, based on IoT and the Raspberry Pi system, allowing the recording and logging of environmental measures in a cloud-based application (Vidva et al. 2025). DIY systems offer significant appeal due to their lower cost and broader customization options compared to limited commercial solutions (see Chap. 9). However, open-source projects may face abandonment or lack of ongoing support due to the unpredictable nature of academic research trajectories.

Generally, any study, especially those utilizing HCM technologies, can benefit greatly by also implementing sensors for illumination, temperature, humidity, noise, pressure, and a passive infrared sensor to acquire precise timestamps of personnel entrances. These sensors have good durability, do not require frequent calibration, and can produce time-series data that can be correlated with other in-cage events, such as home cage activity (Virag et al. 2024; Hu et al. 2023).

Affordable, high-quality sensors like the Sensirion SHT45 (temperature/humidity) and ams OSRAM TSL25721 (light) are readily available, enabling accurate and low-maintenance environmental monitoring at the cage level, which can control for a major source of variability. Sensors for gases like carbon dioxide, ammonia, and volatile organic compounds (VOCs) are available (€60–€300) but often need maintenance. Some, like the Telaire T6713 for CO2, offer long, calibration-free lifespans using infrared dispersion. While useful for ventilated cages, their effectiveness in open-top cages is questionable due to airflow variations.

2.9 Acoustic Monitoring (USV)

Integrating ultrasonic vocalization (USV) recording into HCM systems would significantly enhance our understanding of rodent communication, social behavior, and emotional states. Rodents communicate via USVs (18–110 kHz), which are inaudible to humans. These vocalizations are context dependent, with variations in frequency, duration, and modulation reflecting different emotional states like anxiety, pain, or positive social interactions. While visual monitoring has advanced, USV interpretation has lagged. Incorporating audio monitoring would provide researchers with a more comprehensive behavioral repertoire for assessing animal welfare and stress responses (Portfors 2007; Lefebvre et al. 2020). Consequently, the detection and analysis of USVs have become a marker of animal welfare and an indirect indicator of the internal state of the subject.

USV detection in HCM requires specialized microphones with a broad, ultrasonic frequency response. Captured audio is analyzed using advanced software to identify and categorize call types based on spectral and temporal features. Commercial systems like Avisoft, Dodotronic, and Sonotrack provide dedicated solutions that can be integrated with HCM setups. Similarly, platforms such as those offered by METRIS, Noldus (e.g., UltraVox XT), and BatSound extend these capabilities by combining audio data with other behavioral measures. Additionally, using a 'bat microphone' (Petterson M500-USB, https://batsound.com/product-category/usbmicrophones/) and open source recording software (e.g., Audacity®) is also a viable and significantly cheaper solution to record USVs in laboratory settings.

Integrating ultrasonic audio recording into HCM requires sensitive microphones (18–110 kHz) to convert acoustic waves into electronic signals. Software must synchronize audio capture with other sensor data. Advanced algorithms and machine learning filter noise, segment recordings, and classify call types, automating analysis for affective states and social interactions (Coffey et al. 2019; de Chaumont et al. 2021). Data, audio, video, telemetry, and sensor inputs are integrated to create a comprehensive view of animal behavior, enhancing assessment reliability.

2.9.1 Applications in Behavioral and Welfare Studies

The integration of ultrasonic vocalization detection within HCM systems has several valuable applications. (i) USVs are integral to understanding the dynamics of rodent communication during group housing. Analysis of these vocalizations can elucidate the nuances of social hierarchy, mating behavior, and affiliative interactions (D'amato 1991; Egnor and Seagraves 2016). (ii) Changes in vocalization patterns often indicate stress responses. For instance, alterations in call frequency or intensity may signal discomfort or distress, enabling researchers to assess the impact of experimental conditions on animal welfare (A. Venkatraman et al. 2024). (iii) In drug discovery and testing, monitoring USVs provides an additional behavioral endpoint to evaluate the efficacy and side effects of novel therapeutics. Indeed, subtle shifts in vocalization profiles may reveal the impact of pharmacological agents on central nervous system function (Brudzynski 2015; Simola 2015; Premoli et al. 2023). (iv) Tracking the evolution of USVs over time can offer insights into the developmental trajectory of communication skills in rodents, contributing to studies on neurodevelopmental disorders (Castellucci et al. 2018; Granata et al. 2022).

2.9.2 Pros of USV Systems

- *Non-Invasive Monitoring*: Audio recording is entirely noninvasive, ensuring that animals are not disturbed by the monitoring process.
- *Additional Behavioral Context*: USV data complement other monitoring modalities, such as video or telemetry, by providing an auditory dimension to behavioral analysis.
- *Real-Time Data Collection*: Continuous audio recording enables the capture of spontaneous and transient vocalizations, which might otherwise be missed in periodic assessments.
- *Enhanced Welfare Assessment*: Monitoring USV patterns can provide an early warning system for stress and discomfort, enabling timely interventions to improve animal welfare.

2.9.3 Cons

- *Background Noise Interference*: A major challenge is the potential for background noise, both environmental and equipment-generated, to interfere with the detection of ultrasonic signals.
- *Varying recording quality*: Microphone orientation and distance relative to the animal are crucial. Minor obstructions, like plastic houses, can significantly attenuate USV intensity.
- *Automated Analysis Challenges*: Despite software advances, parsing complex vocalizations and distinguishing individual animals in groups remains challenging.

- *Amount of Data*: Recording ultrasound vocalizations at high sampling rates generates vast datasets, even over short periods, creating real-time analysis and storage challenges. This necessitates limiting continuous observation or incurring higher costs for storage and high-performance technology.
- *Cost Considerations*: High-quality ultrasonic microphones and associated data acquisition systems can be expensive, particularly when multiple recording channels are needed.
- *Individual Identification Limitations*: Single-microphone audio recording alone cannot reliably identify vocalizing animals, unlike RFID or video. This remains a significant limitation, as multimicrophone systems and complex computational methods still struggle with USV emitter identification and incur high costs (de Chaumont et al. 2021; Waidmann et al. 2024). To clearly link vocalizations to individuals, recordings are usually made individually or in specific arenas. However, this often removes the social group and home cage context.

2.9.4 Integration and Future Directions

Integrating ultrasonic vocalization detection into HCM systems is a significant advance in behavioral phenotyping. Future efforts will likely focus on improving audio resolution and developing more sophisticated, real-time analytical algorithms. Emerging technologies may include

- Enhanced Signal Processing: Algorithms that better differentiate between overlapping calls and effectively filter out background noise, possibly using deep learning methods.
- Multimodal Data Fusion: Closer integration of audio data with video and telemetry outputs, allowing for a synchronized analysis of vocalizations alongside movement, physiological parameters, and environmental factors.
- Wireless and Miniaturized systems: As with telemetry, there is ongoing development of smaller, wireless audio recording systems that can be easily incorporated into home cages without interfering with animal behavior.
- Wearable devices with custom circuit boards that can be fitted to individual mice and connected to a variety of existing USV recording systems are currently under development and reached an accuracy of 90% when attributing vocalizations to a specific animal, based on the relative amplitude differences (Waidmann et al. 2024).
- Cloud-Based Data Analysis: Leveraging cloud computing to process and store large volumes of audio data, facilitating large-scale studies and collaborative research efforts.

Audio technology in HCM offers insight into rodent communication, enhancing our understanding of social behavior, stress, and animal welfare. By capturing USVs, researchers gain a rich dataset that complements traditional measures, advancing behavioral neuroscience.

Integrating ultrasonic vocalization detection enriches HCM experimental tools, providing an additional data layer that, combined with other monitoring, offers nuanced interpretations of animal behavior. As technology evolves, improvements in sensitivity, data integration, and analysis will solidify audio monitoring as a fundamental component of home-cage behavioral studies.

2.10 Operant Technology in the Home Cage

Operant systems not only observe *but also directly impact animal* behavior, coupling sensors that detect an animal's voluntary action with actuators that deliver cues, rewards, or punishments to elicit a response. The interaction can be described by three consecutive one-way interactions: (i) the device presents *a cue* to the animal, (ii) the animal exhibits *a response* detected by the device, and (iii) the device provides *a consequence* (reward or punishment) to the animal. The specific behaviors a device can elicit and measure in user-defined operational protocols depend on the electronic components the system is equipped with. Integrating this closed-loop system into the social cage transforms it into an interactive environment for continuous learning and decision-making with minimal human intervention.

2.10.1 The Cue

Cues activate animal senses via electronic components. Components offering wider range and finer control enhance protocol design flexibility. For example, addressable RGB LEDs offer precise light control for varied cues, unlike incandescent bulbs. Protocols must account for animal sensory limits (e.g., mice hear 2–70 kHz, humans 20 Hz-20 kHz).

2.10.2 The Response

A sensor, the main electronic component, translates physical phenomena into electronic impulses, recorded as data. Thus, nearly any sensing technology in this chapter could apply, depending on the behavior scientists aim to recognize as a response to a presented cue.

- Video detection (see Sect. 2.1) uses simple algorithms for presence/movement. Neural networks can also recognize behaviors or body parts; the latter, combined with algorithms (e.g., distance-based interaction), classifies behavior. Real-time neural network inference makes this technically complex.
- RFID tagging (see Sect. 2.2) can be used to detect the presence of one specific animal in a defined area, which can, for example, trigger the start of a test.
- Capacitive sensors can be used to detect touch or licking and function on the same principle as electromagnetic field systems (see Sect. 2.3).
- Infrared motion sensors can be used to detect an animal's presence in an area, while beam-break sensors are commonly used to detect nosepokes in a designated opening (see Sect. 2.4).

- Similar to electromechanical microswitches, load cells and pressure sensors (see Sect. 2.5) can be used to detect an animal pressing on a lever or for area presence detection.
- Ultrasonic vocalization (see Sect. 2.9) can be detected to elicit a consequence. A specific type of vocalization could be filtered using digital signal processing (DSP) algorithms, with or without a machine learning classifier.

Skinner boxes, originally using electromechanical levers, now primarily employ nosepoke openings with beam-break sensors. Levers for rats need to be low and wide for activation, as thin, high levers only elicit sniffing. Nosepokes are better for isolating individual responses in multianimal settings. Ideally, the initial action (e.g., sniffing a nosepoke, standing on a platform) should be neutral before reinforcement or punishment.

2.10.3 The Consequences (Reward Versus Punishment)

Operant conditioning *rewards* often use automated food/water dispensers, with liquid rewards requiring more maintenance. Servo-controlled gates allow diverse reward types. Animal preferences vary, and sugars can complicate experiments. *Punishment*, using air puffs or mild electric shocks, yields consistent reactions and avoids food or water deprivation issues but raises ethical concerns due to stress.

2.11 Benefits of the Home Cage-Based Approach

Conventional operant conditioning is time-consuming (1 month for 1-h daily sessions), limiting its use in adolescence or disease models. For example, assessing ADHD-related impulsivity in rodents is restricted to adulthood due to long training. Home cage setups drastically reduce training time (e.g., <7 days for 5-CSRTT), enable 24/7 testing with many trials (Bruinsma et al. 2019), and may improve learning in a familiar, human-free environment. Shorter training and automation make water gating a more viable reward. Scheduled unattended testing, e.g., 3-h sessions, is possible. Ongoing work hints that food/water deprivation may not be necessary to induce operant learning, at least in rats (Augier et al. 2017). Home cage operant conditioning offers increased statistical power, reduced experimenter workload, and allows testing younger animals and across circadian cycles.

2.11.1 Cognitive Domains Assessed

Operant conditioning protocols assess attention and impulse control (5-CSRTT training in <7 days), cognitive flexibility (IntelliCage reversal/rule-switch protocols), and motivation/effort cost (progressive-ratio hoppers/nose-poke panels). Sensory discrimination and sequence learning are evaluated using PsiBox (tone/frequency discrimination within 2 weeks).

2.11.2 Home Cage Implementation

Configurable home cage operant systems allow for various testing protocols, but translating standard operant tasks to the home cage is not a 1:1 process. For example, a rat home cage 5-CSRTT task showed different impulsivity results and more performed trials when testing was restricted to 2.5 h versus 24/7. These findings highlight new aspects of executive function and the need to reoptimize protocols for the home cage.

2.11.3 Data Acquisition Architecture in Operant Systems

Operant conditioning devices, particularly home cage systems, require precise synchronization and prolonged unattended operation, necessitating robust measures against data loss. While common in embedded systems and automation through redundancy and failure detection, commercial HCM operant systems often rely on a single PC, creating a point of failure. Open-source systems like FED3 (Matikainen-Ankney et al. 2021) offer more resilience by logging to SD cards, but simultaneous, redundant logging and networked architectures with remote alerts would be ideal for preventing data loss and enabling real-time monitoring and control, as demonstrated by nonoperant HCM systems like Tecniplast DVC and MIROSLAV (Virag et al. 2024). Integrating local storage with network logging would further enhance reliability. Given the severe consequences of data loss in animal research, increased adoption of such preventative measures is ethically crucial as HCM systems become more complex.

2.11.4 Pros

- 24/7 Sampling: hundreds of trials across the circadian cycle boost statistical power.
- Voluntary, self-paced interaction keeps stress hormones low and eliminates the experimenter effect.
- Faster acquisition: learning complete in days rather than weeks.
- Group scalability: high-throughput pipelines exist (www.micecraft.com).
- Systems integration: operant event clocks aligned with telemetry, video pose, or environmental sensors enable comprehensive phenotyping.

2.11.5 Cons

- Equipment in contact with animals more prone to malfunction:
  - Dust or bedding on lick spouts: false lick detections and corrupted data.
  - Cable chewing: data loss or hardware failure.
- More severe consequences of equipment malfunction:
  - Servo miscalibration on pellet hoppers: unintended food restriction.
  - Power outages: unsignaled deprivation of food or water and loss of data.

Open-source dashboards bundled with devices like *LIQ-HD* display live lick counts, so anomalies are detected without disturbing the cage (Petersen et al. 2023). Weekly calibration scripts, version-controlled task code, and hardware health checks are recommended best practices.

2.11.6 Implementation Notes for Researchers

When designing an operant HCM cage, link the cue (LEDs/speakers), action (nose-poke/lickometer), and reinforcer (pellets/water) to specific hardware. For cues, use spectrally distinct LEDs or calibrated speakers. For actions, nose pokes localize responses, while lickometers measure precise contact but need water scheduling; combining both distinguishes trial initiation from choice. For reinforcers, use optical gates for pellet verification and daily purging for water lines. Scheduling tasks during the initial dark phase maintain motivation without deprivation, as seen in the 5-CSRTT "time-restricted" protocol.

2.11.7 Data Handling and Analysis

Operant events (subject ID, timestamp, and code) are appended to the HCM log and can be aligned with video pose or telemetry to relate internal state to behavior. A single mouse generates ≈1 MB day^{-1} of event data; automated scripts that bin trials by light phase or stimulus difficulty reduce file sizes and feed dashboards for rapid QC.

3 Combinations of Systems, Emerging Technologies, and Future Considerations

Modern HCM research is moving from single-modality "black boxes" toward interoperable ecosystems in which multiple sensors, actuators, and analytical pipelines talk to each other in real time. This section surveys the technical and conceptual shifts that enable that transition, from hardware miniaturization to metadata standards, and it discusses how ethical and logistical constraints shape the next generation of platforms.

3.1 Aspects to Consider When Driving Innovation in HCM Technologies

- *Miniaturization and Efficiency*: Advancements in nanotechnology and materials science are leading to smaller, more efficient sensors that can be easily integrated into the homecage environment. These sensors can measure a wider range of parameters with greater accuracy and lower power consumption.
- *Integration with Big Data and AI*: Combining sensor data with advanced analytics and machine learning algorithms allows for deeper insights into animal behavior and physiology. Cloud computing and data storage solutions facilitate the management and analysis of large datasets generated by HCM systems.
- *Wearable Technologies*: Flexible electronics and wearable sensors offer the potential for noninvasive, continuous monitoring of physiological parameters in freely moving animals. These technologies can be integrated into lightweight, comfortable devices that minimize stress and behavioral disruption.

- *Personalized and customized monitoring systems*: The development of modular, open-source HCM platforms allows researchers to tailor monitoring systems to their specific research questions and experimental needs. This flexibility enables greater innovation and collaboration within research.

3.2 Key Drivers of Innovation

Advancements in nanofabrication and low-power ASIC design enable sugar-cube-sized multisensor nodes to record locomotion, temperature, and ECG for weeks using harvested energy. These devices, paired with cloud-based GPU clusters, transform home cages into "data lakes" for machine learning to uncover subtle phenotypes. Additionally, open-source modular chassis allow labs to build custom rigs, speeding up iteration and reducing costs (see Chap. 9).

3.3 Retrofitting Versus Green-Field Set-Ups

Where budgets or animal-facility regulations limit new infrastructure, solutions such as Olden Labs' cage-lid modules (https://oldenlabs.com/) or the iMouse retrofit camera system (https://imouse.info/) allow researchers layer monitoring onto standard shoebox cages and commercial racks (Tecniplast, Allentown, …). Conversely, fully integrated arenas like MouseVUER (Salem et al. 2024), LMT (de Chaumont et al. 2019), TSE's Phenomaster (Gallage et al. 2024), or Noldus Phenotyper 2 (Grieco et al. 2021) trade initial capital expenses for richer 3D viewpoints, environmental control, and simplified cabling—at the cost of lower throughput per square meter. Choosing between these extremes depends on the study's spatial footprint, biosafety class, and anticipated life-span of the experiment.

3.4 Embedding Neuroscience Toolchains

Optogenetic stimulators, wireless headstages for calcium imaging, and miniaturized photometry fibers can now be docked inside the home cage so that brain activity, behavior, and environmental cues are recorded synchronously. Proof-of-concept work with 2D-Neuro's wireless optogenetic implants allows freely moving mice to receive pattern-locked stimulation within their home cage settings (https://2dneuro.com/).

3.5 Computer Vision and AI-Based Systems

In recent years, artificial intelligence (AI) has become a cornerstone of next-generation HCM systems, enabling fully automated, high-throughput analysis of complex behavioral and physiological data. By integrating AI with the tools, data, and sensors used in HCM, researchers can extract richer insights, reduce human bias, and accelerate discovery. In the following sections, we describe how AI advances across two primary tasks (identification and interpretation), the evolving role of transformer-based architectures, multicamera 3D solutions, and the trade-offs inherent in edge versus cloud computing deployments.

3.5.1 Identification Task: From Frames to Quantitative Data

The identification task converts raw video frames into numerical representations—animal centroids, body-part keypoints, or object masks—that feed downstream analyses. Early breakthroughs came from repurposing human-pose architectures: DeepLabCut repackaged DeeperCut into an end-to-end toolkit for animal researchers, democratizing markerless pose estimation with just dozens of labeled frames (Insafutdinov et al. 2016; Mathis et al. 2018). Building on this, (Lauer et al. 2022) introduced multianimal pipeline enhancements for faster training and real-time inference, and SLEAP demonstrated robust multianimal tracking in complex arenas (Pereira et al. 2022).

Complementary to pose estimation, object detection networks such as YOLO have been adopted for rapid cage-wide animal localization (Redmon et al. 2015). YOLO's single-pass architecture allows real-time processing on consumer-grade GPUs, making it ideal for large cage racks where dozens of video streams must be monitored simultaneously. Together, these CNN-based tools provide a modular "plug-and-play" identification layer that can be tailored to specific HCM tasks—locomotion quantification, enrichment interaction, or social proximity mapping—while maintaining minimal hardware requirements.

3.5.2 Semantic Segmentation and the Transformer Revolution

While pixelwise semantic segmentation has historically lagged in HCM—due to the laborious annotation of object polygons—recent transformer-based models are poised to change this landscape. The Segment Anything Model (SAM) delivers zero-shot segmentation of previously unseen objects, allowing rapid adaptation to new cage configurations or lighting conditions without manual mask labeling (Kirillov et al. 2023). By prompting SAM with minimal cues, researchers can obtain precise object boundaries—critical for measuring fine-scale interactions with manipulanda or nest materials—without incurring prohibitive annotation costs. This shift from convolutional to transformer architectures (Vaswani et al. 2017) also underpins novel self-supervised methods that learn from vast amounts of unlabeled video. In the near future, HCM platforms may employ such models to discover latent environmental or postural features—beyond predefined keypoints—that correlate with physiological events, further enriching the scope of home-cage analytics.

3.5.3 Three-Dimensional Tracking: Multicamera Integration

Extending AI-driven identification into three dimensions enhances the fidelity of behavioral measures—especially for activities like vertical rearing or climbing. Libraries such as OpenCV have long offered undistortion and stereo-triangulation tools (Bradski 2000), but specialized pipelines streamline 3D pose estimation in rodents. DeepLabCut 3D integrates per-camera keypoints and performs triangulation to reconstruct 3D trajectories (Nath et al. 2019). Building on this, Anipose unifies multicamera calibration and triangulation workflows, simplifying setup across three or more viewpoints (Karashchuk et al. 2021).

Advanced approaches like DANNCE leverage geometric reasoning to resolve occlusions and improve volumetric reconstructions (Dunn et al. 2021), while s-DANNCE introduces scalability optimizations for denser camera arrays (Klibaite et al. 2025) and allows for better social behavior discrimination. Despite the clear benefits of 3D data—precise kinematics and enriched behavioral grammar—the increased hardware complexity and per-cage cost mean that most commercial HCM systems still favor single-camera setups, reserving 3D rigs for specialized studies.

3.5.4 Interpretation Task: From Keypoints to Behavior

Once quantitative features are extracted, the interpretation task employs AI to translate these time series into meaningful behavioral or physiological categories. Supervised classifiers—including support vector machines and deep sequence models—map pose trajectories or raw video clips to ethologically defined behaviors (e.g., grooming, digging, and rearing). While powerful, these methods demand extensive labeled datasets and are limited to prespecified behaviors visible to human observers.

To address annotation bottlenecks and observer bias, unsupervised segmentation algorithms have gained traction. B-SOiD clusters high-dimensional pose dynamics to segment behavior sequences without labels and then derives compact classifiers for real-time use (Hsu and Yttri 2021). Similarly, VAME leverages variational autoencoders to learn latent representations of movement motifs (Luxem et al. 2022), and BehaviorFlow statistically compares behavioral state transition matrices that can be based on any discrete segmentation of behavior, although typically are based on clustering of time-series-embedded gait features (von Ziegler et al. 2024). More recently, Keypoint MoSeq integrated motion sequencing with hidden Markov models to automatically discover hierarchical behavioral modules (Weinreb et al. 2024). These unsupervised methods not only reveal novel patterns—such as subtle gait alterations preceding motor deficits—but also reduce reliance on human-defined ethograms, improving reproducibility.

3.6 Ethical Considerations

- Minimizing Stress and Discomfort: Homecage monitoring systems should be designed and implemented in a way that minimizes stress and discomfort for the animals. Noninvasive or minimally invasive technologies are preferred whenever possible.
- Impact of Monitoring Devices: Researchers should carefully assess the potential impact of monitoring devices on animal behavior and welfare. This includes considering the size and weight of devices, their placement on the animal, and any potential for interference with normal activities.

- Regulatory Compliance: All homecage monitoring studies must adhere to relevant ethical guidelines and regulations for the care and use of laboratory animals.
- Balancing Data Needs with Animal Welfare: Researchers must carefully balance the need for scientific data with the welfare of the animals. Open and transparent communication with ethical review boards and the public is essential for maintaining public trust and support for animal research.
- Be careful to not reduce welfare and limit the translational and physiological relevance of the extended amount of data recorded by trying to integrate too much at a time. Welfare should be a priority, not just a set of rules constraining the development of the technologies.

3.7 Interoperable Metadata Across HCM Systems

As described in Chap. 11, it is more than critical that systems and different technologies are able to communicate efficiently with each other. In addition, they should provide standardized and fully interoperable set of metadata schemas. It is therefore highly recommended that both academics developing DIY solutions (Chap. 9) and manufacturers work together toward a consensus for (meta)data management as recently proposed and described (Bains et al. 2025). For example, when companies develop new technologies, they should now prioritize the interconnectivity of their solutions with already existing systems. This could be simplified by developing and offering open APIs that could allow scientists to integrate and combine their variations in system databases. Offering TTL communications is also a great way to facilitate the use and communication between systems, but integrating interoperability at the software level should become a reflex as well. For example, individual solutions are currently being developed with the goal of offering a unified metadata management system, integrating, standardizing, and unifying metadata at the level of project elaboration, mouse management, and data publication (Metadatapp (MAPP), www.metadatapp.net).

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